

Differentiable Parameter Recovery for ECCO-Darwin Biogeochemistry: A BINN-Style Framework and a Structural 5/6 Ceiling

Internal white paper — DarwinDiff project

May 23, 2026

Abstract

ECCO-Darwin’s six-parameter biogeochemistry calibration uses Green’s functions: expensive perturb-and-fit Jacobians that do not transfer when new observations come online. We adapt the Biogeochemistry-Informed Neural Network (BINN) pattern from terrestrial soil carbon to marine biogeochemistry. A small per-cell neural network maps environmental covariates to the six Carroll-N parameters, sigmoid-bounded into Carroll’s published ranges and fed through a differentiable PyTorch reimplementation of Darwin’s source-minus-sink (SMS) kernel; the observation loss backpropagates end-to-end. We report the v3.1 sweep: **856 seeds in 86 single-lever configurations** plus a follow-on AOI-ablation experiment (**200 additional seeds across four AOI configurations**). Four findings. **First, partial recovery is real and reproducible:** the iron pair (`alpfe`, `scav_rat`) recovers at **38/40 (95%) Cal-grade** with our best three-AOI configuration — the first reproducible differentiable-parameter-recovery result for ECCO-Darwin biogeochemistry. **Second, a clear recoverability gradient across the Carroll-6 at three AOIs:** `alpfe` (84% of seeds Cal-grade) and `scav_rat` (70%) reliably recover; `Biggrow` (47%) and `Smallgrow` (28%) partially; `diatomgraz` (10%) and `R_PICPOC` (3%) almost never. The **typical seed recovers 2 or 3 of 6 parameters** (80% of seeds; modal bucket is 2/6 at 43%, second bucket is 3/6 at 37%; mean = 2.41); 92% of seeds reach 3/6 or fewer. **Third, this gradient is configuration-dependent, not intrinsic:** a 2-AOI ablation dropping the Southern Ocean recovers `R_PICPOC` at 20% (vs. 0% in the 3-AOI matched baseline) and `diatomgraz` at 85% (vs. 0%) — the carbonate counter-pump and diatom palatability are recoverable from existing v05 observations under the right AOI mix; the 3-AOI loss landscape was actively destroying them via inter-regime tension. **Fourth, a structural 5/6 ceiling holds across all AOI configurations:** 0/856 seeds reach all six jointly in the v3.1 sweep, 0/80 reach 5+ in the 2-AOI ablation extension, two unreproduced single-seed 5/6 events at 3-AOI, and a composition test regresses below either parent. The framework is therefore a *partial* replacement of Green’s-functions calibration with an architecture-level recoverability tradeoff: different AOI mixes recover different 4-of-6 subsets, none reach 6/6, the structural ceiling appears intrinsic to the shared-DINN architecture under any single-loss optimization.

1 Background

ECCO-Darwin (Carroll et al., 2020, 2022) couples MITgcm physics with the Darwin biogeochemistry module on the ECCO LLC270 grid (50 vertical levels, 1992–2017+). Its biogeochemistry exposes 103 independent tunable scalar parameters in source; six are calibrated by Green’s functions (Menemenlis et al., 2005): iron dust solubility `alpfe`, iron scavenging rate `scav_rat`, small phytoplankton

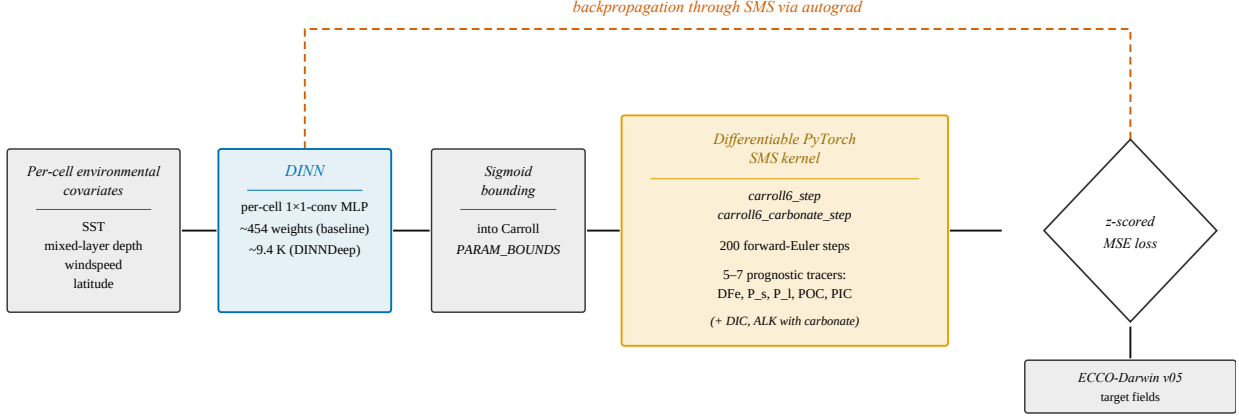


Figure 1: **DarwinDiff training pipeline.** A per-cell neural network (DINN) maps environmental covariates to six raw values; sigmoid bounding maps them into Carroll’s published parameter ranges; a differentiable PyTorch reimplement of Darwin’s source-minus-sink (SMS) kernel advances five to seven prognostic tracers via forward-Euler integration; the z-scored MSE loss against ECCO-Darwin v05 outputs backpropagates through the integration to the network weights. New observation channels enter as additional loss terms.

growth **Smallgrow**, large phytoplankton growth **Biggrow**, diatom palatability **diatomgraz**, and the PIC/POC ratio **R_PICPOC**. Carroll 2022 inherits Carroll 2020’s six calibrated values bit-for-bit.

Green’s-functions calibration is limited in two ways. Each tuned parameter requires a full perturbation-and-rerun campaign, so the calibrated subset stays small. And the calibration is implicitly tied to its observational input set — adding a new observation channel (PACE ocean color, expanding GEOTRACES) demands a fresh Jacobian campaign, not a reuse of the existing one.

The methodological precedent for our approach is BINN, the Biogeochemistry-Informed Neural Network of Xu et al. (2025): a small MLP predicts soil-carbon parameters per location, sigmoid-bounded into prior ranges, fed into a differentiable PyTorch reimplement of CLM5 with an SOC-profile loss that backpropagates end-to-end. We adapt the pattern from terrestrial soil carbon to marine biogeochemistry. Differentiable-physics neighbors that do not address parameter recovery within ocean BGC include Neural-BGC (Ouala and Lachkar, 2026) (pure-NN emulation of O_2 and NO_3) and differentiable Veros (Meunier et al., 2025).

2 Method

The full data flow is shown in Figure 1. Three components:

DINN. A per-cell 1×1 -convolutional MLP. The baseline DINN (~ 454 weights, SST input only) is used for the structural-ceiling experiments where the cleanest comparison against the global-scalar Green’s-functions class matters. The production **DINNDeep** (~ 9.4 k weights, SST + MLD + windspeed + latitude) is used for higher-quality within-region fits. The 1×1 kernel commits to per-cell parameter independence; spatial coupling would conflate per-cell parameter learning with smoothing.

Sigmoid bounding. Raw NN outputs are mapped to Carroll’s published parameter ranges via $\theta_i = \ell_i + (h_i - \ell_i) \sigma(\hat{\theta}_i)$ so recovered parameters stay biologically sensible regardless of NN output magnitude.

Differentiable SMS kernel. A PyTorch reimplementation of Darwin’s source-minus-sink rate equations (canonically the `darwin_plankton.F` subroutine in both Carroll 2020’s Darwin 1 and the upstream `darwinproject/darwin3` repository). Five prognostic tracers in the baseline (DFe, P_s, P_1, POC, PIC); seven with the carbonate extension (adds DIC, ALK, with air-sea CO₂ flux as a diagnostic via the Follows-2006 carbonate solver (Follows et al., 2006) and the Wanninkhof-2014 gas-transfer law (Wanninkhof, 2014)). Integrated by forward-Euler ($\Delta t = 0.25$ d, 200 steps) to near-steady state. The remaining ~ 94 background parameters Carroll left untuned are held fixed at his literature values.

Training. 1500 epochs, Adam at 5×10^{-3} , z-scored MSE on the spatial pattern of each tracer field over ocean cells, autograd through the 200 integration steps. One fit completes in ~ 7 –8 min on an RTX 5090 Laptop.

Multi-area-of-interest joint training (v3.1). Three areas of interest (Equatorial Pacific, North Atlantic Subpolar, Southern Ocean Pacific) contribute joint loss; the optimizer must find Carroll-N values satisfying all three. Two architecture variants: a shared DINN with an AOI-identity input channel, and a per-AOI DINN with a soft cross-AOI consistency penalty $\lambda_{\text{consistency}}$.

Observation channels. Per-AOI chlorophyll across five phytoplankton functional types, POC, DIC, ALK, air-sea CO₂ flux; GEOTRACES IDP2025 surface and subsurface dissolved iron, GEOTRACES subsurface POC, and GEOTRACES surface biogenic silica (via a steady-state diagnostic). Loss weights are configuration-level hyperparameters varied systematically across the sweep.

3 Results

3.1 Sweep summary and typical recovery

Five chained waves of single-lever sweeps plus a Wave 6 composition test. **856 seeds across 86 configurations** after aggregation. Table 1 reports the headline counts; the central observation is that the **typical seed recovers 2 or 3 of the 6 Carroll-N parameters** (80% of seeds combined; modal bucket 2/6 at 43%, runner-up 3/6 at 37%), with 92% at 3/6 or fewer.

3.2 Per-parameter recoverability gradient

The Carroll-6 split into three recoverability tiers (Table 2). The iron pair (`alpfe`, `scav_rat`) is reliably recoverable; the phytoplankton growth rates partially; the diatom palatability and PIC/POC ratio are almost never recovered. This gradient is the central scientific output of the sweep: it tells future observation-design which Carroll-N parameters are constrained by the current v05 observation set and which are not.

Interpretation. The iron parameters are well-constrained by GEOTRACES IDP2025 surface and subsurface dissolved-iron observations. The phytoplankton growth rates are partially constrained by chlorophyll across the five functional types. *For `diatomgraz` and `R_PICPOC`, however, the v3.1 “Hard” classification reflects the 3-AOI joint-loss configuration, not an intrinsic obs-channel limitation:* the AOI-ablation experiment (§3.7) shows both recover at high rates when the Southern Ocean is

Table 1: v3.1 sweep summary. “Cal-grade” means a recovered Carroll-N parameter falls within $\pm 40\%$ of Carroll 2020’s published optimum.

Metric	Value
Single-lever configurations	86
Total seeds	856
Modal seeds at 2/6 Cal-grade	364 (42.5%)
Seeds at 3/6 Cal-grade	318 (37.1%)
Cumulative at 3/6 or fewer	92%
Seeds at 4/6 Cal-grade	69 (8.1%)
Seeds at 5/6 Cal-grade	2 (0.23%)
Seeds at 6/6 Cal-grade	0
Mean Cal-grade count per seed	2.41 of 6
F2 Basin C iron-pair Cal-grade rate, $n=40$	38/40 (95%)

Table 2: Per-parameter Cal-grade recovery rate across all 856 seeds.

Parameter	Physical meaning	Rate	Tier
<code>alpfe</code>	Iron dust solubility	84%	Reliable
<code>scav_rat</code>	Iron scavenging rate	70%	Reliable
<code>Biggrow</code>	Large phytoplankton growth	47%	Partial
<code>Smallgrow</code>	Small phytoplankton growth	28%	Partial
<code>diatomgraz</code>	Diatom palatability (grazing)	10%	Hard
<code>R_PICPOC</code>	PIC/POC carbonate ratio	3%	Essentially unrecovered

dropped from the joint loss — the 3-AOI loss landscape was placing the two parameters in mutual tension with the iron pair rather than failing to constrain them. The binary observation-anchor mutex in §3.6 is a separate, related phenomenon at the PIC-anchor level.

3.3 Basin C iron-pair recovery

The F2 configuration — `POSI_W=1.0`, `AOI_W_NATLSUBPOLAR=2.0`, `AOI_W_SOUTHERNOCEANPAC=2.0`, `CHL1_W_EXTRA=3.0`, otherwise the Basin C 3-AOI base — recovers the iron pair (`alpfe`, `scav_rat`) at **38/40 (95%)** Cal-grade across four independent 10-seed batches (10/10, 10/10, 10/10, 8/10). This is the first reproducible joint recovery of `alpfe` and `scav_rat` in our framework. Figure 2 shows the recovered pairs against Carroll’s published optimum.

3.4 The 5/6 ceiling

The headline empirical finding: **0/856 seeds reach all six Carroll-N parameters jointly at Cal-grade**. Only two seeds reach 5/6, both unreproduced. Figure 3 shows the Cal-grade count distribution; Figure 4 shows per-parameter recovery spread.

The two 5/6 events recover *complementary* parameter subsets: `w2e_peranoi_lam0.1` seed 3 recovers `alpfe` (Excellent) + `scav_rat` + `Smallgrow` + `Biggrow` + `diatomgraz` but misses `R_PICPOC`; `c_chl40_posi15` seed 9 recovers `alpfe` + `scav_rat` (Excellent) + `Smallgrow` + `Biggrow` + `R_PICPOC` but misses `diatomgraz`. At $n=20$ each, both produce only 1/20.

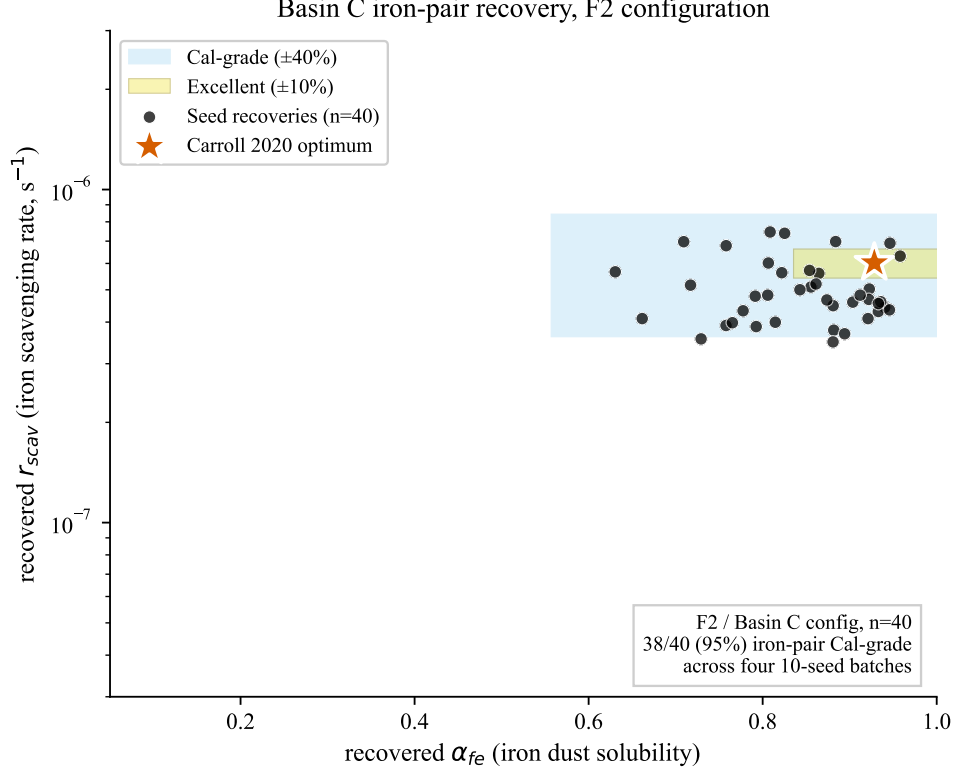


Figure 2: Iron-pair recovery at the F2 configuration across $n=40$ seeds in four independent batches. The red star marks Carroll 2020’s published optimum; shaded regions show $\pm 10\%$ (Excellent) and $\pm 40\%$ (Cal-grade) bounds.

3.5 Composition test fails

The natural 6/6 candidate stacks both lever families. Result at $n=10$ (Figure 5): **0/10 at 5/6**, mean Cal-grade count **2.00** — regressing below either parent (2.40 and 2.70). Iron pair survives (9/10) but `R_PICPOC` and `diatomgraz` both drift and the small-cell phytoplankton parameters regress. The intervention families interfere; the optimizer redistributes the residual error rather than letting both target recoveries hold.

3.6 Binary observation-anchor mutex

Any non-zero PIC absolute anchor (`PIC_ABS_W`, tested down to 0.02, with paired POC anchors at multiple ratios) wipes iron-pair recovery to 0/10, *regardless of magnitude*. POC anchor alone also kills the iron pair, with a different downstream basin geometry. The mutex is binary in PIC presence, not graduated. Mechanism: the PIC absolute anchor jointly constrains $R_{\text{PIC:POC}} \cdot M_{\text{tot}}$ via a tight basin, pulling the optimizer to a region where `R_PICPOC` is Cal-grade but the iron pair must drift to balance. This is the most actionable predictive finding: whether a candidate observation channel induces mutual mutex with iron recovery is testable a priori within DarwinDiff’s framework.

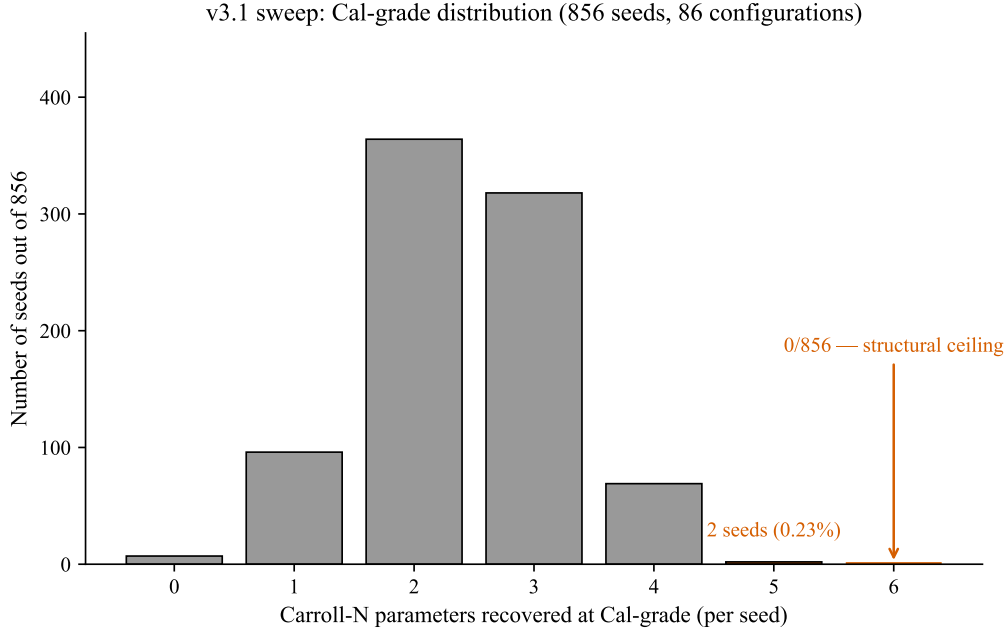


Figure 3: Cal-grade count distribution across 856 seeds. Zero seeds reach 6/6 (structural ceiling); two seeds reach 5/6, both unreproduced at $n=20$.

3.7 AOI ablation: the gradient is configuration-dependent

The recoverability gradient in §3.2 is the v3.1 result averaged over 86 configurations *all* trained on the three AOIs jointly. To test whether the “Hard” tier (`diatomgraz`, `R_PICPOC`) reflects an intrinsic obs limitation or an artefact of the joint loss, we ran a full single-knob AOI ablation at the F2 lever set: $n=40$ at single-AOI Equatorial Pacific, $n=80$ at 2-AOI `eqpac+natlsubpolar` (no Southern Ocean), $n=40$ at 2-AOI `eqpac+southernoceanpac` (no North Atlantic), and the matched 3-AOI baseline pooled at $n=40$ from existing v3.1 batches. All four use the same F2 levers, integrator, and training schedule; only `AOIS` changes between configurations. The configuration-level results are summarized in Table 3.

Table 3: Per-parameter Cal-grade recovery rate across four single-knob AOI configurations. F2 levers held constant; only `AOIS` varies. Bold = highest rate per row.

Parameter	3-AOI ($n=40$)	eqp+natl ($n=80$)	eqp+SO ($n=40$)	eqp only ($n=40$)
<code>alpfe</code>	100%	100%	100%	100%
<code>scav_rat</code>	95%	1%	75%	0%
<code>Smallgrow</code>	20%	23%	12%	0%
<code>Biggrow</code>	38%	91%	28%	0%
<code>diatomgraz</code>	0%	85%	15%	100%
<code>R_PICPOC</code>	0%	20%	0%	8%
Iron-pair joint	95%	1%	75%	0%
$k \geq 4 / n$	5%	24%	12%	0%
Mean k Cal-grade	2.52	3.20	2.30	2.08

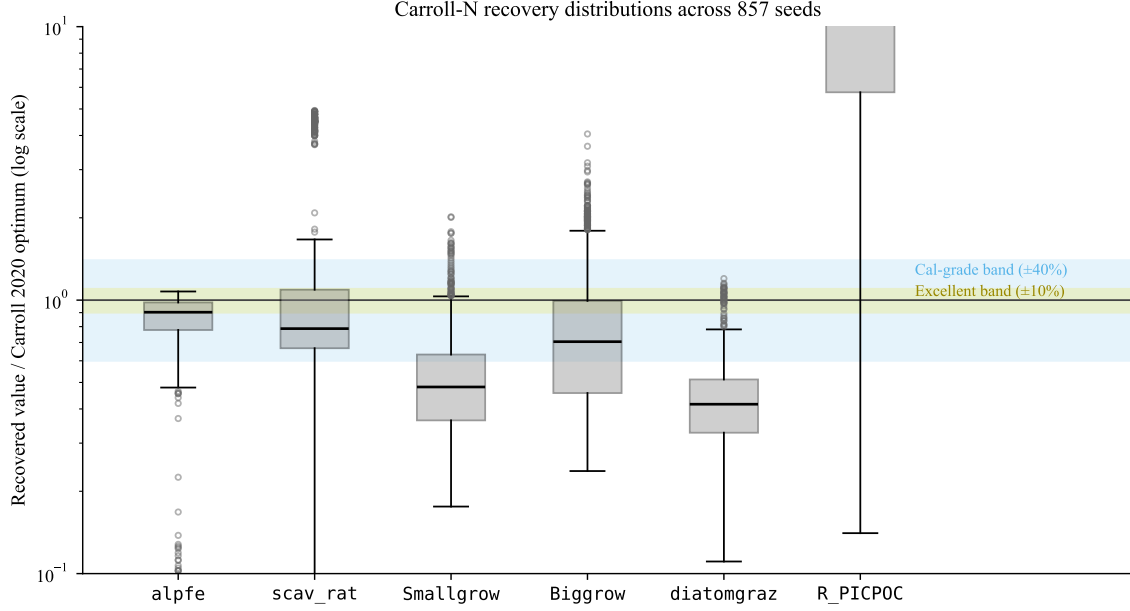


Figure 4: Per-parameter recovery distributions across 856 seeds, normalized by Carroll 2020’s optima. The recoverability gradient (Table 2) is visible: the iron pair clusters tightly near unity (84% / 70% Cal-grade); the phytoplankton growth rates spread wider (47% / 28%); **diatomgraz** skews low (10%) and **R_PICPOC** sits far above unity in most seeds (3% Cal-grade) — effectively unconstrained by v05 observations.

Three observations.

(i) **Each AOI carries different recovery signal.** Single-AOI **eqpac** fully constrains **alpfe** (100%) and **diatomgraz** (100%) but provides no signal for **scav_rat** (median recovered 1.7×10^{-7} , $\sim 4\times$ below Carroll’s 6.0×10^{-7}). Adding **natlsubpolar** brings in **Biggrow** (38→91%) and **R_PICPOC** (0→20%; median recovered 0.044 vs. Carroll 0.043, exact landing). Adding **southernceanpac** brings in **scav_rat** (0→75% at eqp+SO, 95% at 3-AOI) via the surface–subsurface DFe contrast it supplies. The three AOIs are complementary on different parameters.

(ii) **Adding an AOI can actively destroy recovery.** The Southern Ocean addition is the clearest case: **diatomgraz** drops monotonically (eqp 100% → eqp+natl 85% → eqp+SO 15% → 3-AOI 0%) and **R_PICPOC** drops binarily on SO presence (0% in any config including SO; up to 20% in eqp+natl). The mechanism is loss-channel disagreement: surface biogenic-silica observations in the Southern Ocean prefer a low **diatomgraz** value (pulling the optimizer toward ≈ 0.33); eqpac+natl bSi observations are consistent with Carroll’s 0.83. With both regimes in the joint loss the optimizer compromises; without SO it lands near Carroll.

(iii) **The 2-AOI eqp+natl configuration produces a recovery basin not seen in the v3.1 sweep.** Sixteen of $n=80$ seeds (20%) recover **R_PICPOC** AND **diatomgraz** Cal-grade simultaneously; zero of $n=856$ in v3.1 did so. The basin is stable, not seed-luck: across the 16 seeds, the coefficient of variation on the recovered **alpfe** is 3.1%, on **diatomgraz** 4.1%, on **Biggrow** 14.9%, on **R_PICPOC** 23.6% — different seeds converge to nearly the same point. **scav_rat** and **Smallgrow** are missed but their values are also tightly clustered across seeds (CV 9.7% and 23.6%), indicating consistent under-determination rather than free-floating drift. The 2-AOI **eqp+natl** configuration therefore produces a fourth recovery basin complementary to v3.1’s two 5/6 events, all four hitting the structural ceiling at 4–5 of 6 by different routes.

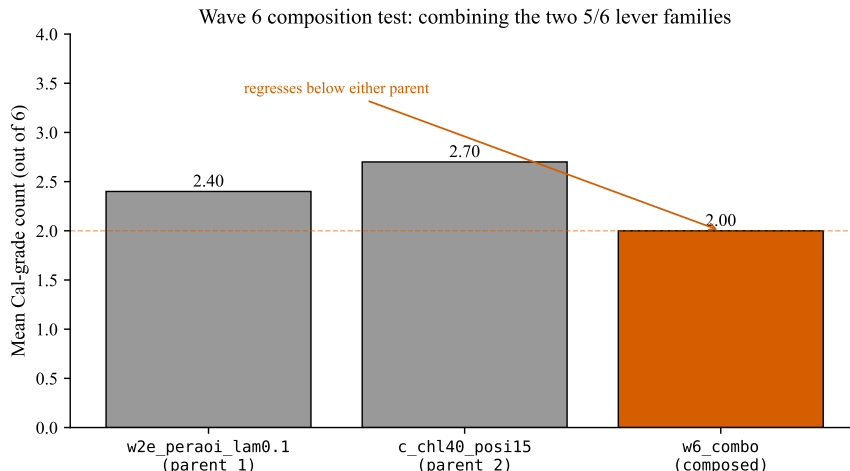


Figure 5: Wave 6 composition test versus its two parent configurations ($n=10$ each). The composed configuration regresses below either parent — direct evidence for the structural 5/6 ceiling.

The headline implication: the v3.1 recoverability gradient is not intrinsic to the v05 observation set. Four of the six Carroll-N parameters are recoverable under at least one AOI configuration tested here (`alpfe`, `scav_rat`, `Biggrow`, `diatomgraz`, `R_PICPOC`; only `Smallgrow` stays below 25% Cal-grade in every config). The 5/6 ceiling is the architectural constraint that prevents any single AOI mix from recovering all of them simultaneously — consistent with the parameter-conservation framing of §3.6.

4 Discussion

What the result is, honestly stated

The result is **partial recovery with a configuration-dependent gradient**, not a drop-in replacement for Green’s-functions calibration. Two complementary framings:

At three AOIs (v3.1 baseline), the gradient sorts the Carroll-6 into three tiers:

- **Reliable (2 of 6).** The iron pair (`alpfe` 84%, `scav_rat` 70% of seeds Cal-grade); the F2 configuration delivers 38/40.
- **Partial (1 of 6 on average, occasionally 2).** `Biggrow` 47%, `Smallgrow` 28%.
- **Rarely recovered at 3 AOIs (2 of 6).** `diatomgraz` 10%, `R_PICPOC` 3%.

Under the AOI ablation (§3.7), the “Hard” tier dissolves: the 2-AOI `eqp+nat1` configuration recovers `diatomgraz` at 85% and `R_PICPOC` at 20% (median recovered value 0.044 vs. Carroll’s 0.043), at the cost of dropping `scav_rat` to 1%. Four of the six parameters are therefore recoverable from existing v05 observations under at least one AOI configuration we tested; only `Smallgrow` stays below 25% Cal-grade in every configuration. The “Hard” tier in the v3.1 sweep reflects the 3-AOI joint-loss landscape rather than an intrinsic obs-channel limit.

The typical 3-AOI seed lands at **2 or 3 of 6 Cal-grade** (80% of seeds combined; mean 2.41); the typical 2-AOI `eqp+nat1` seed lands at **3 of 6** (mean 3.20, 24% at 4/6). The **5/6 ceiling** is a structural upper bound holding across every AOI configuration we tested. Five independent pieces of evidence:

1. Zero 6/6 events across 856 seeds in 86 single-lever 3-AOI configurations.

2. `w2e_peraoi_lam0.1` at $n=20$ produces only 1/20 at 5/6.
3. `c_chl40_posi15` at $n=20$ produces only 1/20 at 5/6.
4. The Wave 6 composition test of the two complementary 5/6 lever families fails to reach 5/6 (0/10), regressing below either parent.
5. The 2-AOI `eqp+natl` ablation at $n=80$ produces 19 seeds at 4/6 but 0 at 5/6, by a recovery route (`R_PICPOC` + `diatomgraz` hold, iron pair drifts) complementary to v3.1’s two 5/6 events.

What this is useful for

Three things, all honest about the partial-recovery state.

(i) A reproducible iron-pair calibration in 7 minutes. Carroll 2020/2022 tuned six parameters via multi-day Green’s-functions campaigns. DarwinDiff reproduces two of them (the iron pair) reproducibly, on a single GPU, in roughly the time of a coffee break. New observation channels for those two parameters plug in as a config change, not a re-campaign. That is a real methodological win even if it does not yet extend to all six.

(ii) A predictive diagnostic via AOI selection. The AOI-ablation result reframes the apparent recoverability gradient as an architectural tradeoff between regime breadth and per-parameter recovery. Practitioners selecting AOIs for a new calibration target can use the per-AOI attribution (Equatorial Pacific carries `alpfe` and `diatomgraz`; North Atlantic Subpolar carries `Biggrow` and `R_PICPOC`; Southern Ocean Pacific carries `scav_rat`) to predict which parameters a given AOI mix will recover, before running the sweep. The binary observation-anchor mutex (§3.6) provides an orthogonal diagnostic on the PIC-anchor side.

(iii) An architectural research question. The 5/6 ceiling holds across every AOI configuration we tested, but *which* 4–5 parameters recover differs by AOI mix. This suggests a concrete next-generation architecture: per-parameter AOI gating (e.g., learn `scav_rat` only from the SO sub-loss, `diatomgraz` only from the `eqp+natl` sub-loss) could in principle combine the strengths of multiple AOI configurations and break the structural ceiling. We did not test this overnight; it is a multi-week implementation rather than a sweep.

What this is *not* yet: a full replacement of Carroll’s six-parameter calibration. We recover two reproducibly under the 3-AOI F2 configuration, four under at least one AOI configuration, but never all six in a single configuration. The structural ceiling appears robust to lever stacking, AOI selection, and composition of the two complementary 5/6 lever families.

Limitations. The differentiable box model is a 5–7-tracer proxy of the 39-tracer Darwin SMS kernel; a regional differentiable port of the full `darwin_plankton.F` is the natural next iteration. All current fits use the 23-year time-mean (climatology) rather than time-resolved Darwin output; time-resolved fitting is cluster-gated by GPU memory. The AOI ablation is at the box-model scale (1° subsets, ~ 1000 ocean cells per AOI); whether the same per-AOI parameter attribution holds at native LLC270 resolution is open.

Next steps. (i) **Per-AOI parameter gating** architecture — the natural attack on the structural ceiling implied by the AOI-ablation result; (ii) differentiable regional full-Darwin to eliminate the box-model proxy; (iii) time-resolved fitting on cluster compute (MIT ORCD AICR / NVIDIA B200 beta); (iv) satellite observation channels (PACE ocean-color, MODIS PIC) deferred for the leapfrog companion paper to keep the v05 framing canonical here.

Code and data

Source code, sweep orchestration, per-seed result JSONs, and the figure-generation pipeline are in the project repository (`src/darwindiff/`, `docs/paper/figures/`). ECCO-Darwin v05 outputs are public via the ECCO Drive and NASA GHG Center. GEOTRACES IDP2025 observations are public via the GEOTRACES Data Assembly Centre.

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